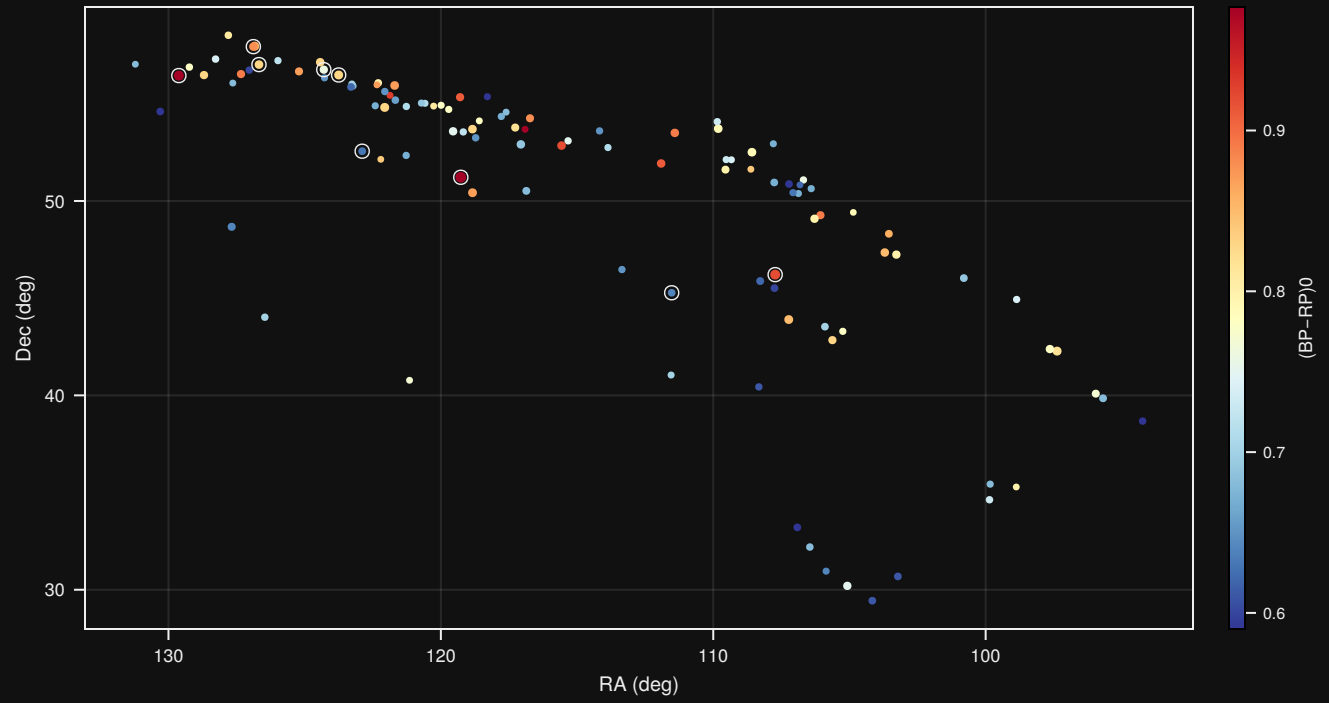
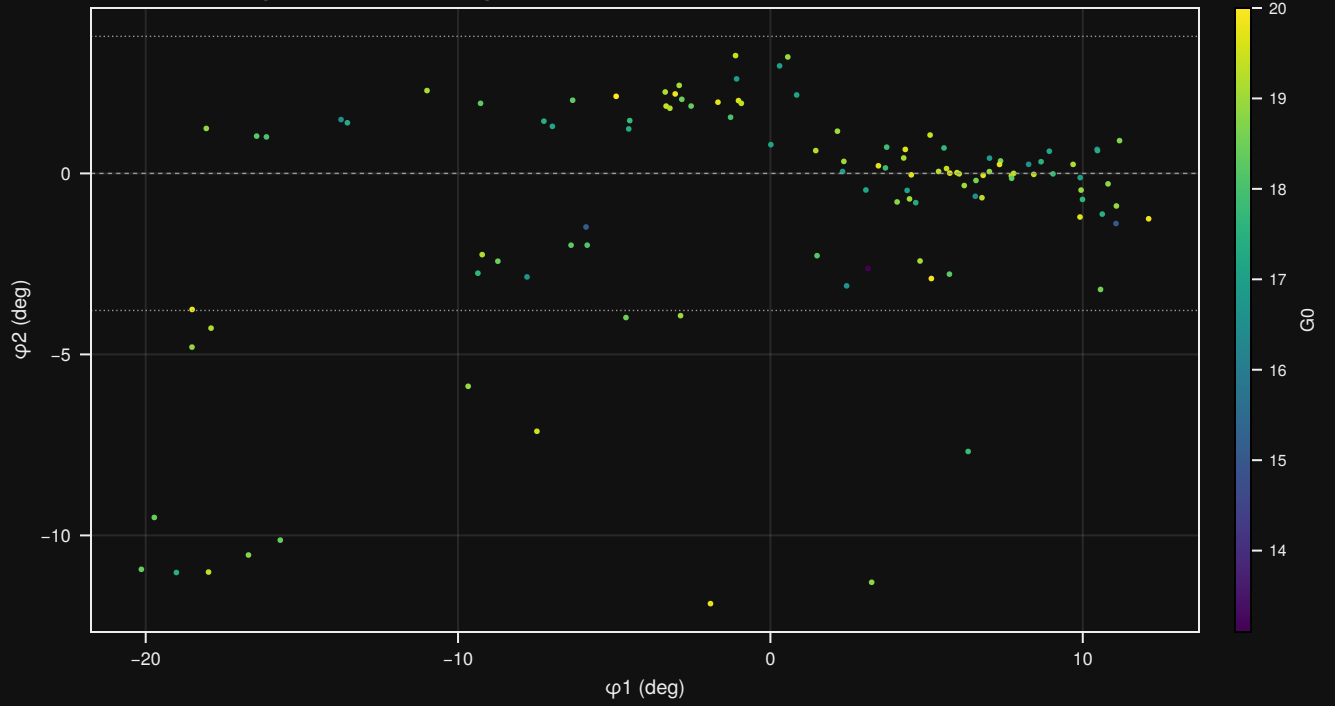
C-11

MEMBERSHIP	
Stars in catalog	112
with Gaia GSP-Phot	55 (49%)
with radial velocity	9
Stream id (sID)	25
GEOMETRY · great-circle fit	
Centroid RA, Dec	114.8, 52.8 deg
Centroid l, b	167.2, 29.7 deg
Pole RA, Dec	340.9, 28.1 deg
Length (φ1, 2–98%)	30.0 deg
Width σ(φ2), robust	1.89 deg
KINEMATICS	
pmRA* med ± σ	6.09 ± 1.03 mas/yr
pmDE med ± σ	-1.84 ± 0.97 mas/yr
μφ1* med ± σ	3.77 ± 1.05 mas/yr
μφ2 med ± σ	-5.28 ± 0.76 mas/yr
Parallax (cat.) med	0.222 ± 0.216 mas
Parallax (wtd, ZP-corr.)	0.271 ± 0.017 mas
Distance	d = 3.7 ± 0.2 kpc
Physical length	1.93 kpc
RV median ± σ(MAD)	-13.5 ± 66.4 km/s
RV gradient dV/dφ1	-4.57 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.1–20.0 (18.7)
(BP−RP)0 median	0.74
GSP-Phot [M/H] (n=55)	-2.03 ± 0.93
GSP-Phot Teff median	5472 K
RUWE > 1.4	0.0 %
PM clusters (largest)	3 (35 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	112 | 112
n_RV catalog | paper	9 | 9
<[Fe/H]> paper	-2.91
Stellar mass (paper)	1100 Msun
Pole offset vs paper	7.9 deg
Remark	Velocities unclear

Sky map · catalog members (n = 112) · size ∝ brightness · ring = has RV



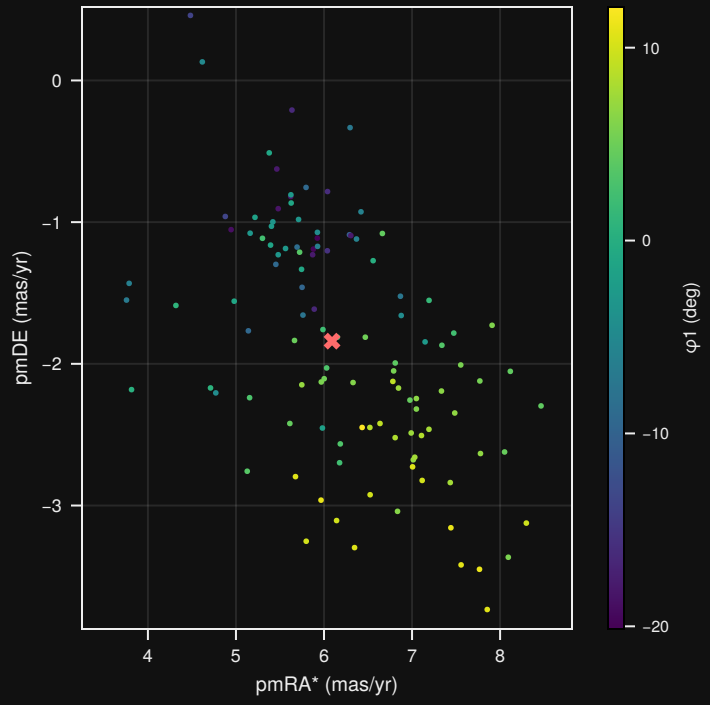
Stream frame (great-circle fit) · length 30.0° · width σ 1.89°



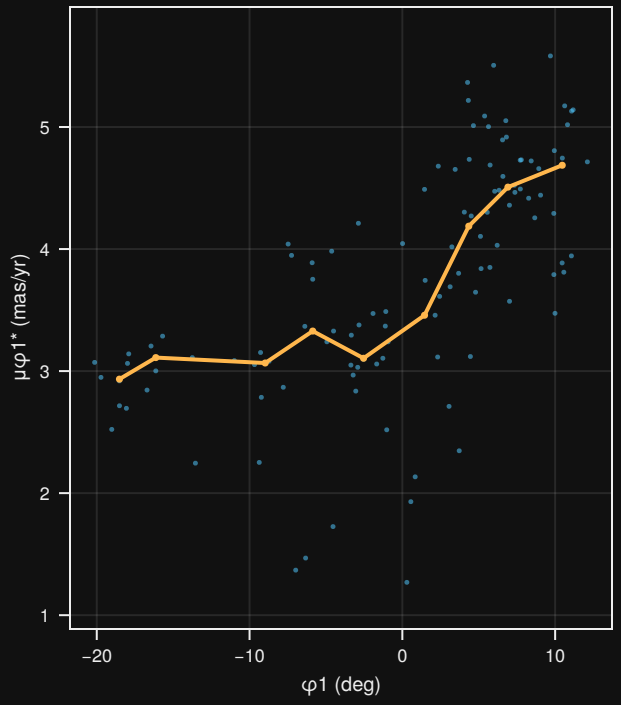
C-11

Kinematics · proper motion · parallax · radial velocity

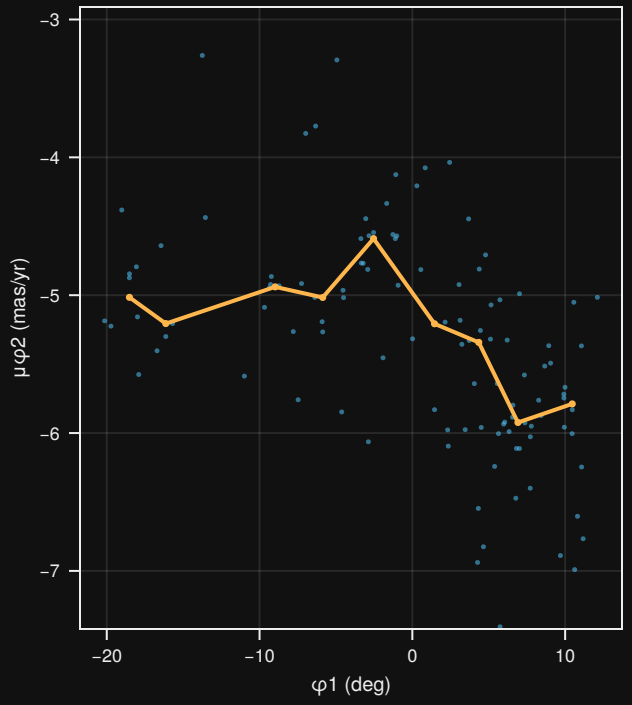
Proper motion (n = 112)



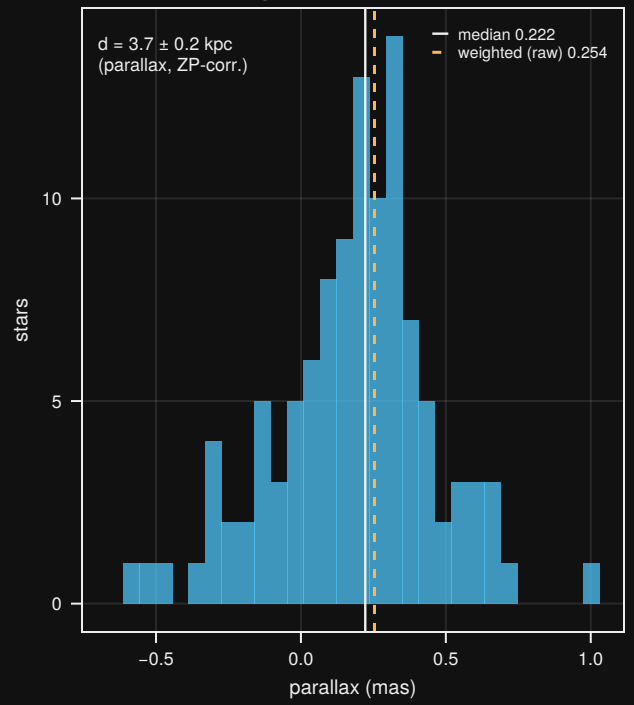
μφ1* vs φ1 · σ = 1.05



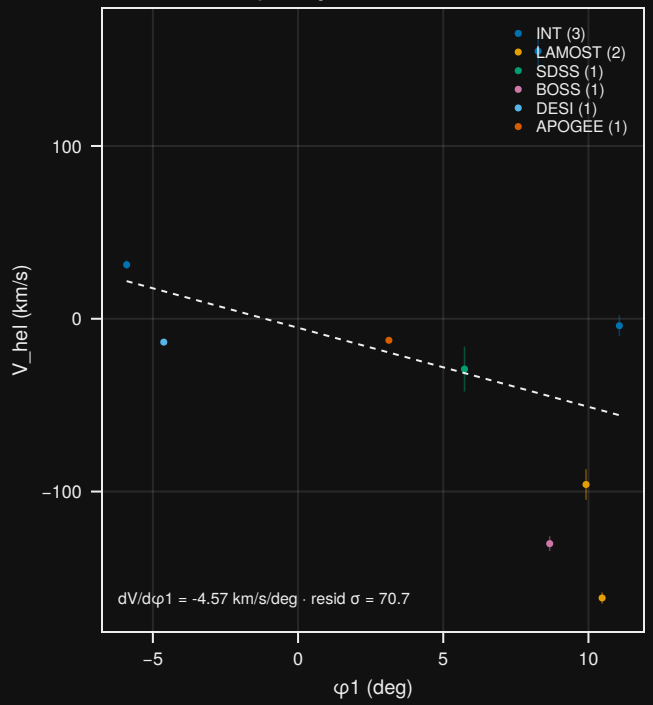
μφ2 vs φ1 · σ = 0.76



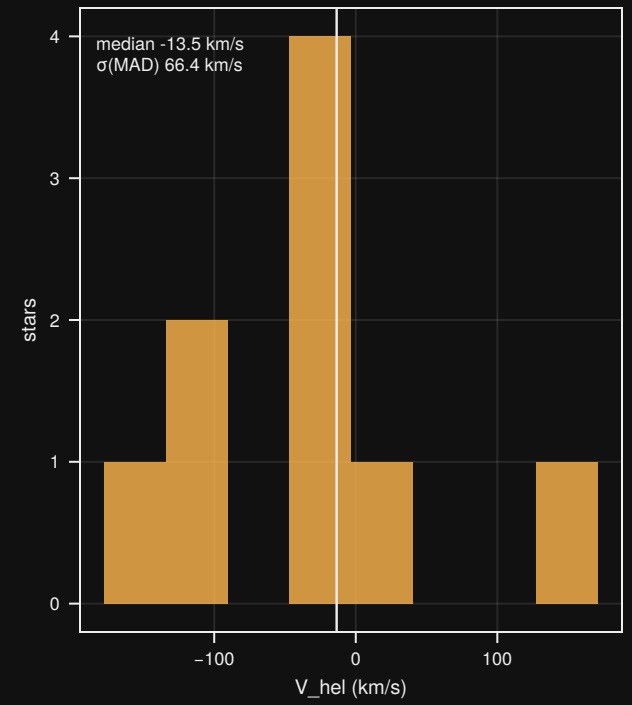
Parallax (catalog, n = 112)



Radial velocity vs φ1 (n = 9)

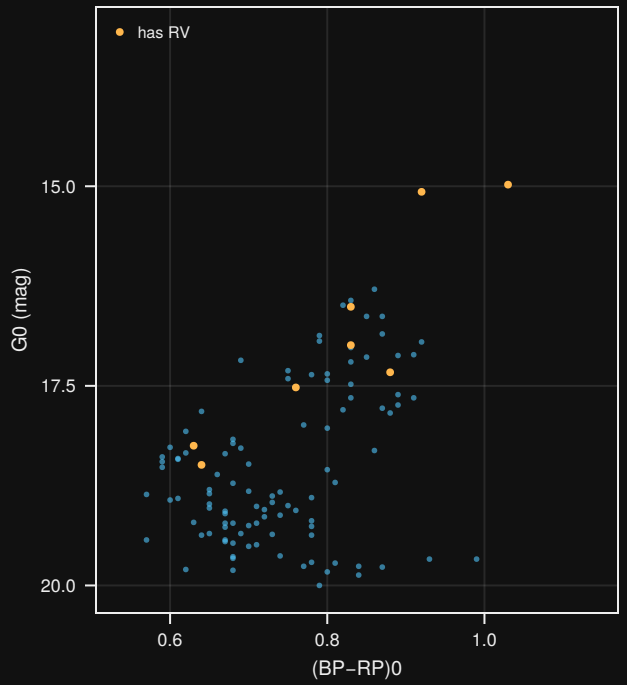


V_hel distribution

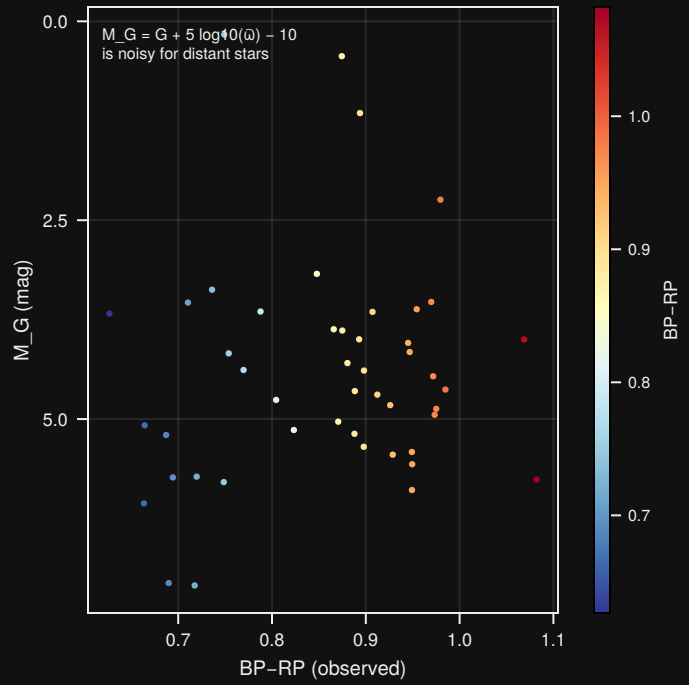


C-11

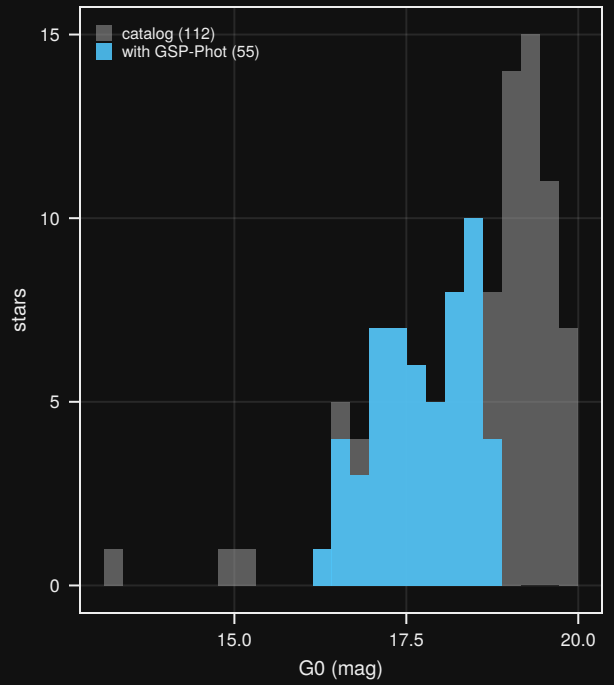
CMD · dereddened (n = 112)



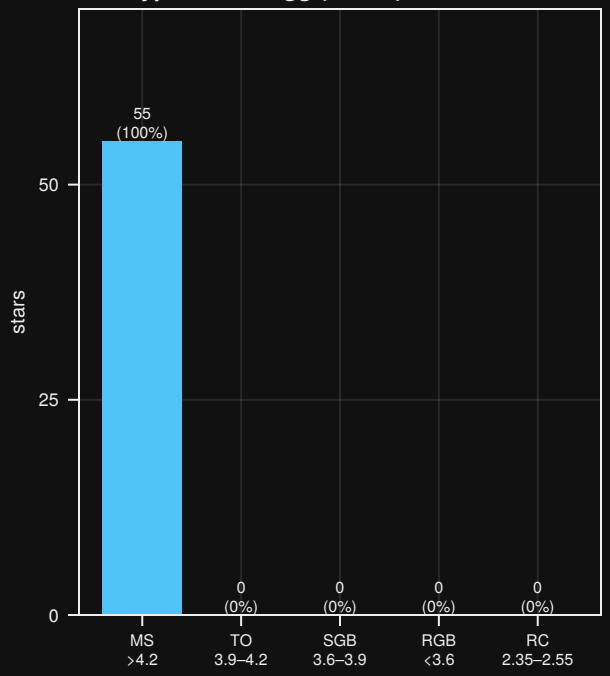
HR diagram · ϖ>0 (n = 47/55)



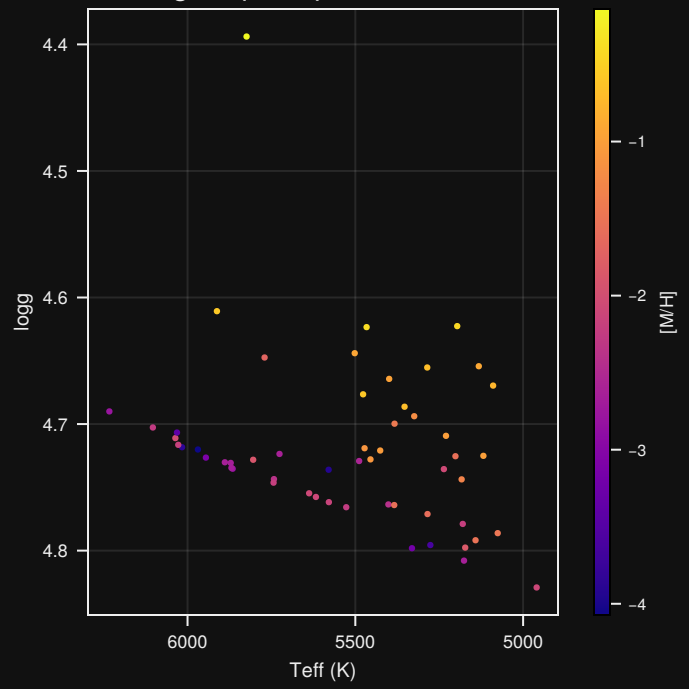
G0 & GSP-Phot coverage



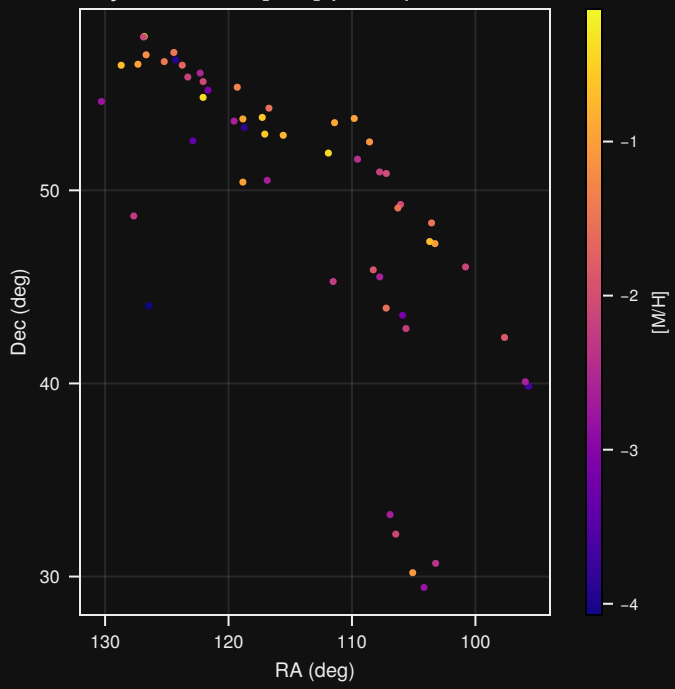
Star types from logg (n = 55)



Kiel diagram (n = 55)



Sky · GSP-Phot [M/H] (n = 55)

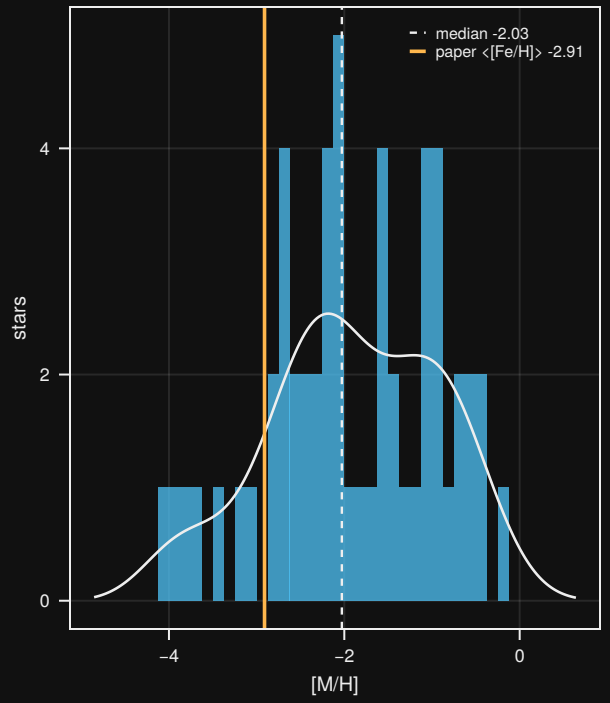


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

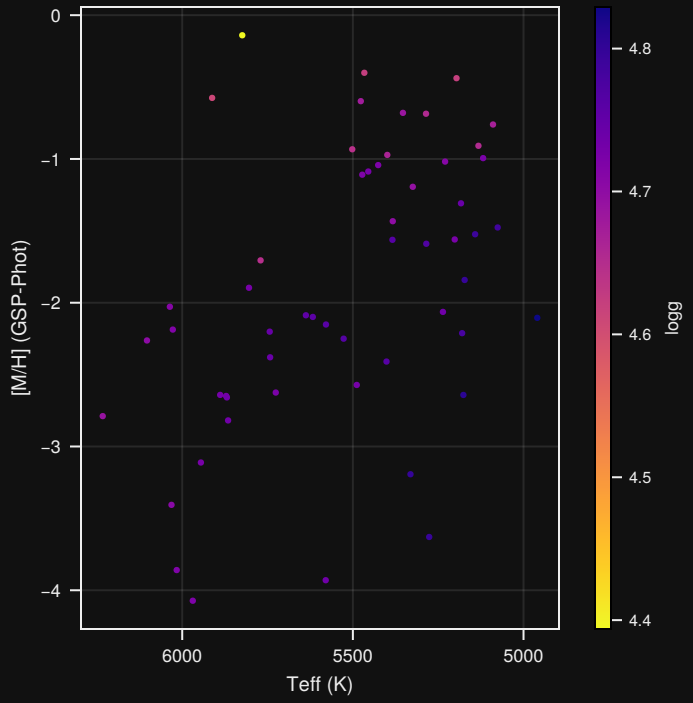
C-11

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 55)



Teff vs [M/H] (n = 55)



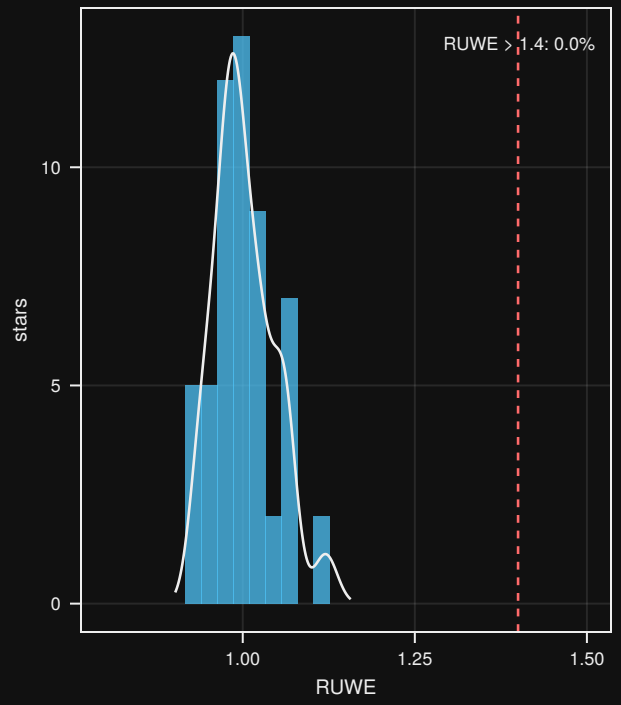
FLAME age (n = 1)

FLAME ages for only 1 star(s)
(FLAME needs good parallaxes)

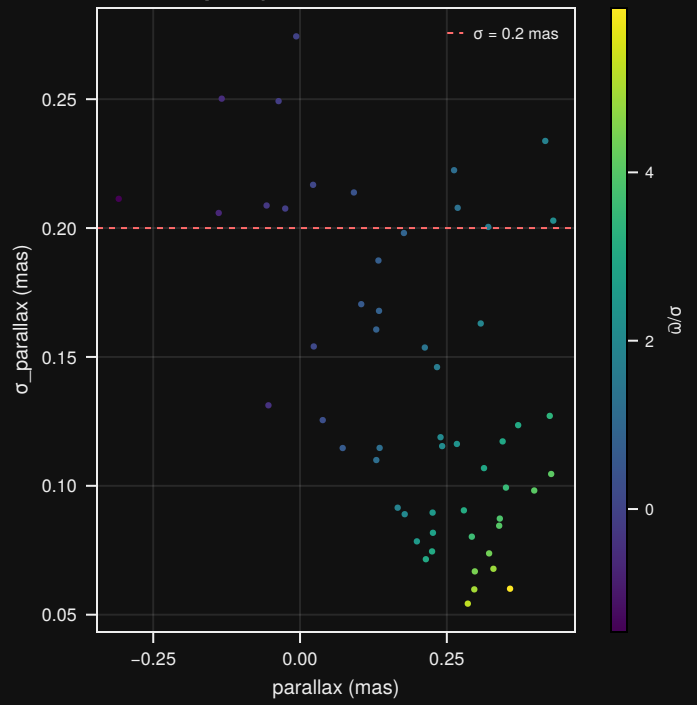
Age–metallicity (n = 1)

n < 3

RUWE (n = 55)

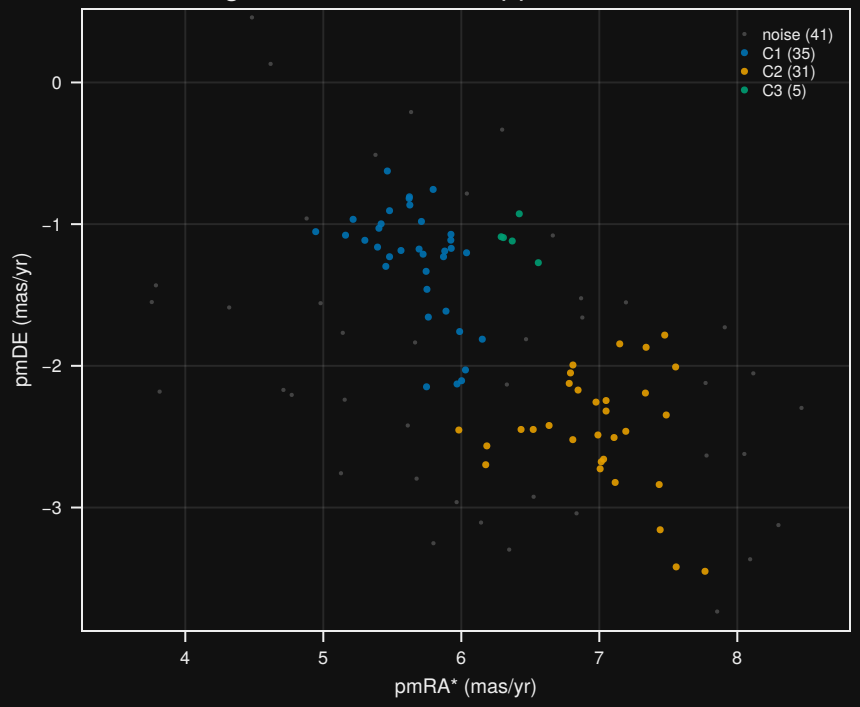


Parallax quality (n = 55)

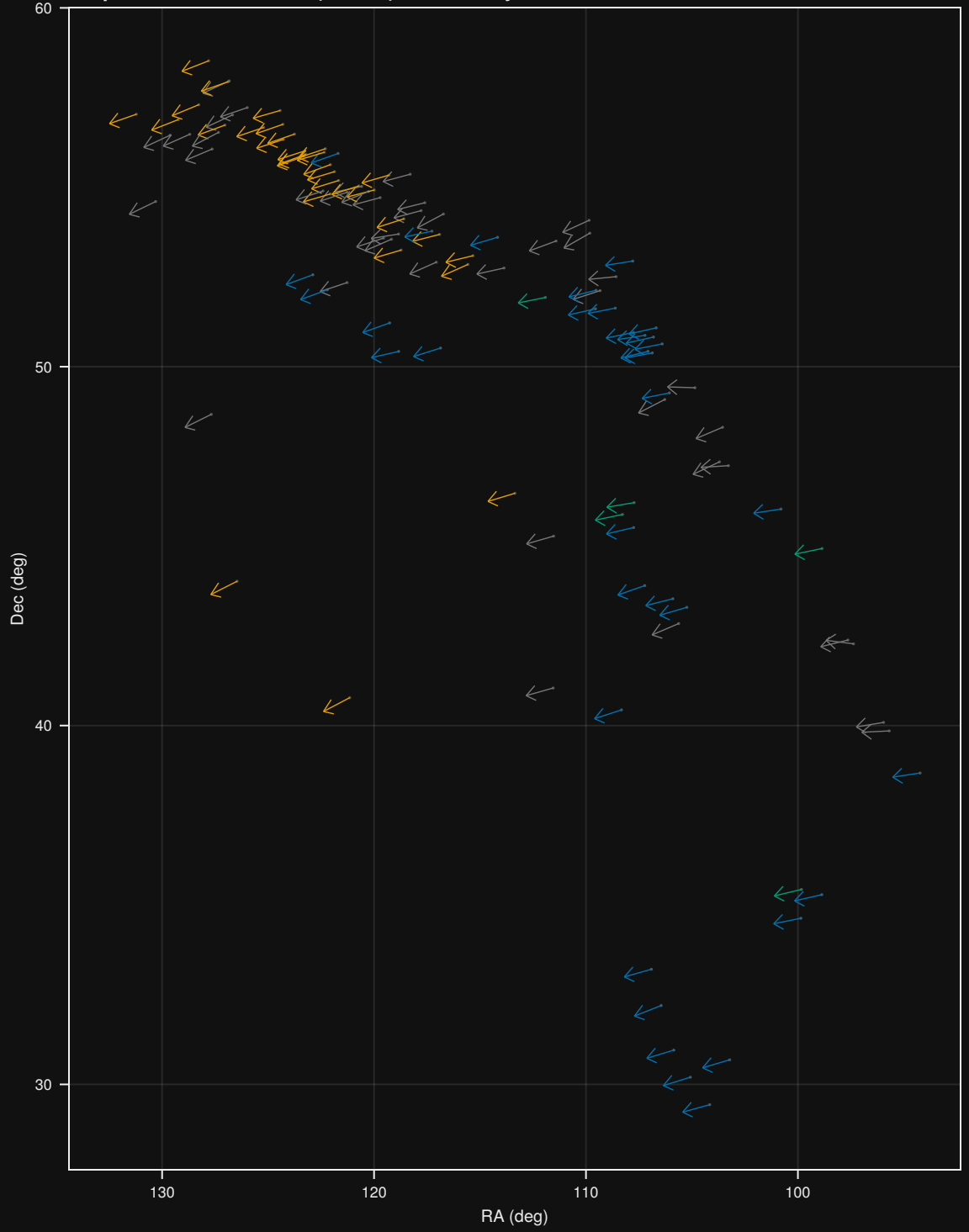


C-11

PM clustering · HDBSCAN · 3 cluster(s) · noise 37%



Proper-motion directions (arrows) coloured by cluster



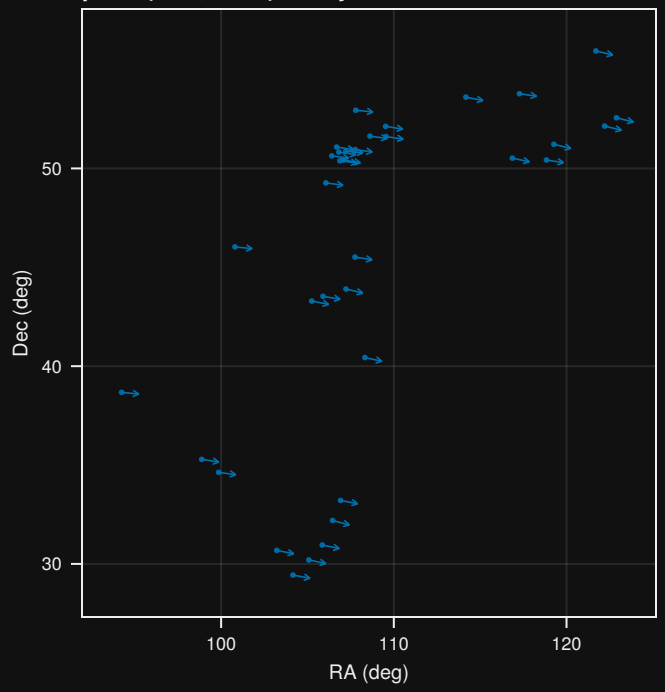
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	41	6.04	-2.12	1.35	-1.36	3
C1	35	5.72	-1.18	0.30	-2.20	2
C2	31	7.03	-2.45	0.37	-2.10	3
C3	5	6.37	-1.09	0.09	-1.17	1

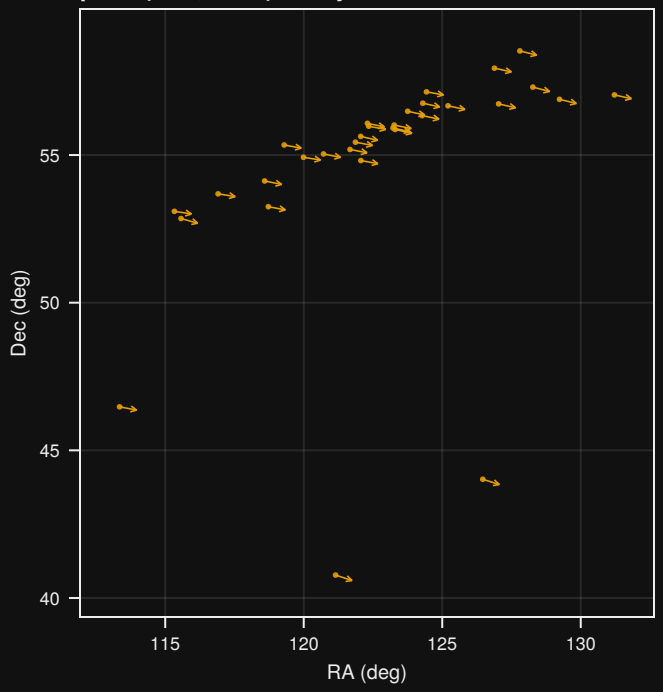
C-11

Per-cluster proper-motion vector maps

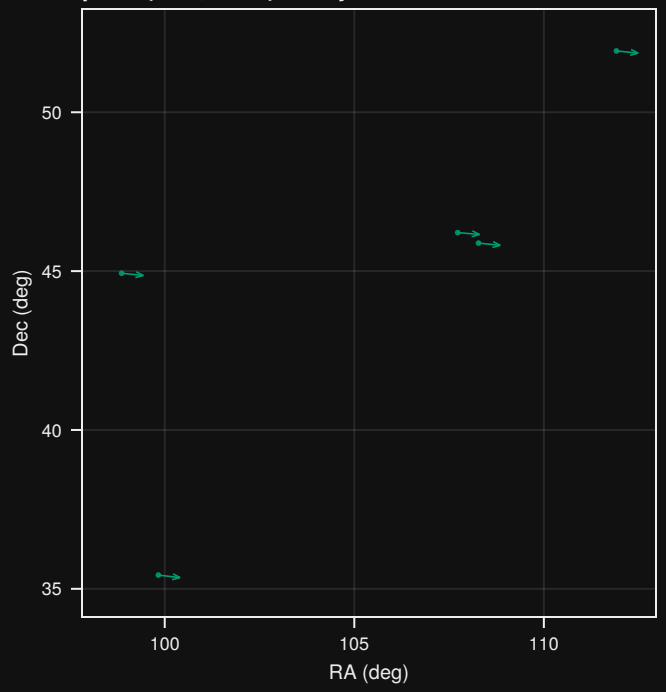
Cluster C1 · n = 35
pm = (5.72, -1.18) mas/yr



Cluster C2 · n = 31
pm = (7.03, -2.45) mas/yr



Cluster C3 · n = 5
pm = (6.37, -1.09) mas/yr



Noise · n = 41

